Journal of Hazardous Materials 499 (2025) 140146

Contents lists available at ScienceDirect

Journal of Hazardous Materials

journal homepage: www.elsevier.com/locate/jhazmat

Biodegradable polylactic acid microplastics enhance methylmercury production in soils and promote mercury accumulation in rice grains in Northeastern China

Zhao Wenwen [a,b], Zhang Zimo [a], Wu Haobo [a,b], Sang Luan [a,b], Zhang Zhongsheng [a,*], Wu Haitao [a]

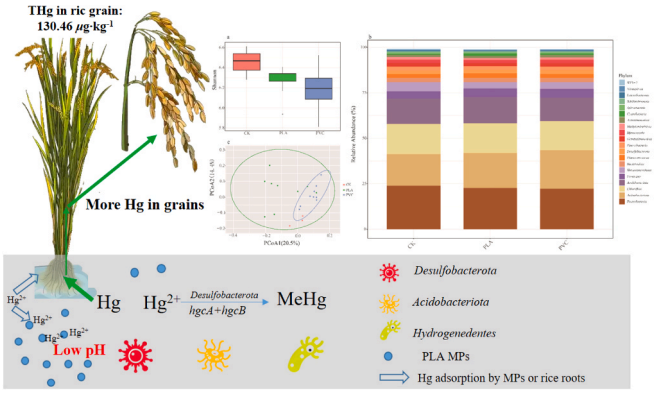
[a] Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun 130102, China
[b] University of Chinese Academy of Sciences, Beijing 100049, China

HIGHLIGHTS

- PLA is more powerful in fueling MeHg production and increasing Hg in rice grains.
- MPs addition had no significant influences on microbial community structures.
- PLA enhance Hg methylation gens expression.
- *Desulfobacterota* was the predominant Hg-methylation taxa across all treatments.
- *hgcA* and *hgcB* positively related to Hg contents in rice grains.

GRAPHICAL ABSTRACT

MPs addition may promote the production of MeHg in soil and prolong mercury retention time in paddy soils, of which PLA is more effective in fueling MeHg formation. Soil pH is greatly changed by PLA addition, and is the important driver of MeHg. *Acidobacteriota* is the key phylum that controls MeHg either in PLA or in PVC treatments. Soil pH and microorganisms primarily regulate soil methylmercury concentrations through indirect effects, particularly by modulating mercury methylation processes via competitive interactions with *Desulfobacterota*. MPs reduced the accumulation of Hg in rice roots by competing with the roots for Hg and adsorbing it. But, translocation of Hg to rice grains were remarkably enhanced by PLA, resulting in high THg contents in rice grains, 130.46 $\mu g \bullet kg^{-1}$ on average.



ARTICLE INFO

Keywords:
Microplastics
Mercury
Methylmercury
Rice grains

ABSTRACT

Microplastics (MPs), as emerging pollutants, have been accumulating in historically Hg-polluted paddy fields. However, the effects of MPs on mercury (Hg) methylation and subsequent accumulation in rice organs have remained poorly understood. We conducted controlled pot experiments to evaluate the impacts of replacing conventional polyvinyl chloride (PVC) with biodegradable polylactic acid (PLA) on Hg methylation processes in soil-rice systems. Results revealed that the addition of PLA-MPs significantly enhanced methylmercury (MeHg)

* Corresponding author.
E-mail address: zzslycn@iga.ac.cn (Z. Zhongsheng).

https://doi.org/10.1016/j.jhazmat.2025.140146
Received 5 May 2025; Received in revised form 28 September 2025; Accepted 14 October 2025
Available online 15 October 2025
0304-3894/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

PVC
PLA
Microorganisms

formation in soil, leading to substantially higher Hg accumulation in rice grains. The average grain Hg concentration reached 130.46 $\mu g \bullet kg^{-1}$ under PLA treatment - approximately 6-folds higher than in the controls. While MPs amendment reduced bacterial α-diversity (Shannon index), it did not substantially alter the overall microbial community composition. Notably, PLA-MPs significantly upregulated Hg methylation gene expression (*hgcA* and *hgcB*). Soil pH was significantly and negatively related to *hgcA* expression. *Desulfobacterota* was the predominant Hg-methylation taxon across all treatments. Importantly, *hgcA* and *hgcB* expression levels showed strong positive correlations with Hg accumulation in rice grains. Our findings emphasize the necessity for comprehensive risk assessment regarding the implementation of biodegradable MPs in Hg-rich agricultural soils, particularly concerning food safety implications stemming from enhanced Hg enrichment in rice grains.

1. Introduction

Rice is the primary staple food for more than half of the world's population, and most people's exposure to mercury (Hg) in China is related to frequent rice consumption [1]. The formation of neurotoxic methylmercury (MeHg) in paddy fields is greatly promoted by the long-term flooded periods (about 3–6 months every year), which create oxygen-deficient soil environment and anaerobic predominant microorganisms. Microbial methylation by sulfate-reducing bacteria [2], iron-reducing bacteria [3], and methanogens is primarily responsible for relatively high MeHg levels in paddy soils [4]. An estimated 1.5–6.6 million tons of MPs are dumped onto and stored in global agricultural soils [5], causing the co-existence of MPs and historical Hg pollutants in paddy fields [6,7]. However, little is known about how MPs affect Hg methylation in paddy soils.

Recent research has revealed that MPs can markably enhance Hg methylation by acting as carriers and promoting Hg accumulation in biota [8,9]. By creating biofilms, MPs adsorb nutrients effectively and provide new niches for microbiota, as termed plastispheres [10], thereby produce more MeHg in paddy soils [11]. Pot experiments confirmed higher *hgcA* expression in "plastisphere" than bulk soils, resulting in 1.7 folds higher MeHg concentrations with MPs addition compared with the control [11]. In contrast, a study that added polyvinyl chloride MPs (PVC-MPs) into red soil and alkaline soil reported MPs reduced MeHg bioavailability by changing dissolved organic matter composition and affecting abundance of *Proteobacteria*, *Firmicutes*, and *hgcA* genes [6]. It remains largely unclear whether MPs promote Hg methylation in paddy fields, which highlighting the need for systematic advancement in our understanding of the relationships between MPs and MeHg production. Crucially, investigations of MPs-MeHg interactions based on incubation experiments must be conducted in soil-rice systems, rather than exclusively in soils. Meanwhile, soil environment, microbial community, Hg methylation genes expression, and MPs biodegradability should be considered to deciphering impacts of MPs on MeHg concentrations and Hg transferring and accumulation in rice organs and grains. However, to the best of our knowledge, no relevant studies have specifically investigated whether the co-occurrence of MPs and Hg pollution promotes Hg accumulation in rice plant organs and grains.

Northeast China experienced a significant expansion of paddy planting areas from 2000 to 2017, reaching 6.25×10^6 ha in 2017 with an average annual increase of 0.20×10^6 ha [12]. Paddy soils of Northeast China are potentially undergoing combined pollution from Hg and MPs, with Hg concentration ranging from 6 $\mu g \bullet kg^{-1}$ to 7770 $\mu g \bullet kg^{-1}$ [13] and with the mean MPs concentration of 270 $\mu g \bullet kg^{-1}$ [14]. Hg levels in agriculture soils are expected to increase due to atmospheric deposition from coal combustion during long winter heating season [15]. The main MPs in arable soils are polyvinyl chloride (PVC), as approximately 50,000 tons of PVC are used for greenhouse film in Northeast China [16]. However, there is a growing movement to replace traditional, fossil-fuel based plastics like PVC with biodegradable alternatives such as polylactic acid (PLA), aiming to alleviate plastic contamination. PLA is more biodegradable, serving carbon and nutrients source for microbes, and is expected to more effective in stimulating Hg methylation. But, more experiments are need to validate this assumption. We hypothesized that MPs could promote MeHg formation in soil-rice systems, and PLA replacing PVC could enhance Hg accumulation in rice tissues and grains. To test this hypothesis, we conducted a 180-day pot experiment with PLA and PVC amendments. Hg and MeHg levels in soils and rice organs (roots, leaves, and grains), bacterial communities (16*S rRNA sequencing)*, and methylate genes expression (*hgcA* and *hgcB*), were analyzed to assess MPs' effects on Hg methylation and accumulation in rice.

2. Materials and methods

2.1. Sample preparation and MPs for pot experiments

The topsoil (0–20 cm depth) was collected from rice fields in Changchun for pot experiments. The soil was air-dried, ground to pass a 2-mm steel sieve, and storage in air temperature before pot experiments. The soil contained 1.52 % of organic carbon (SOC), 0.224 % of nitrogen (TN), 0.129 % of sulfur (S), 416.5 mg/kg of phosphorus (P), and the pH was 8.19. Two MPs powders (PLA and PVC) of 50 μm were bought for pot experiments from commercial company in Guangdong Province.

2.2. Pot experiment design and sample collection

The MPs and Hg contents in pot experiments were determined according to relevant literature. To better elucidate the mechanism by which MPs influence Hg methylation, MPs concentrations are usually deliberately set significantly high, reaching to 0.2–10 % [6,11], which s far above than those found in actual soil environments. In the present study, we set MPs contents to 0.05–0.5 % in pot experiments. Considering Hg levels in arable soils from Northeast China changed in the range of 6–7770 $\mu g \bullet kg^{-1}$ [13], and the extremely high values of 18, 130 $\mu g \bullet kg^{-1}$ found in Huludao City [17,18], a moderate Hg levels, about 3200 $\mu g \bullet kg^{-1}$ THg, was used in pot experiments.

Pot experiments were started at June 20, 2024 and finished at October 12, 2024. About 12.5 kg (dry weight) of the paddy soil was added into iron buckets for each treatment: no MPs addition as the control (CK), PLA addition at 0.2 % and 0.5 % of soil weight, and PVC addition at 0.2 % and 0.5 % of soil weight. Each treatment was replicated three times. Total mercury content (THg) in soils were adjusted to 3200 $\mu g \bullet kg^{-1}$ by adding $HgCl_2$ solution (100 $mg \bullet L^{-1}$). The soils and MPs were thoroughly mixed to achieve homogenization and kept flooded for 7 days before planting rice (Fig. S1).

Five groups of rice seedlings (each group consisting of six seedlings) were planted in iron buckets, placing these iron buckets a greenhouse where the temperature was maintained at 25 ± 5 °C. Urea (15 g) and potassium dihydrogen phosphate (8 g) were applied as basal fertilizers in each pot. An additional 10 g urea was added to each iron bucket during the tillering and heading stages of rice. Water was supplied every 3 days to sustain the flooded condition throughout the growth period, and was drained on September 28, 2023, when the rice reached maturity, in order to enhance the grain plumpness and boost the yield, following typical rice cultivation practices in the field.

Rice plants were carefully collected when pot experiments finished,

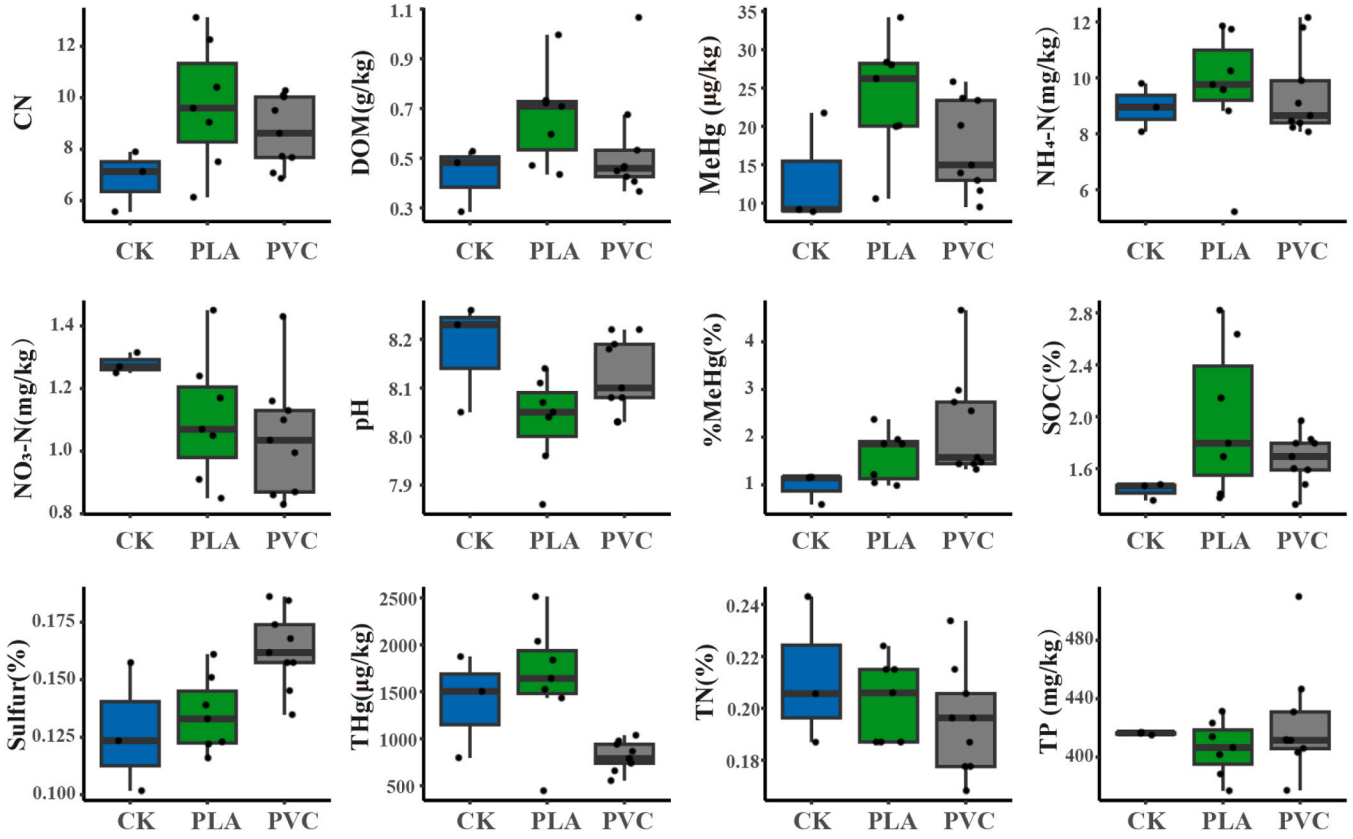


Fig. 1. Comparison of soil properties among soils under CK, PLA and PVC treatments. Multi-comparison results were showed as letters in box plots. The same letters meant no significant difference while the different letters indicated significant difference at $p < 0.05$ level.

and roots, leaves and grains were separated by hand. Leaves were oven-dried for 30 min at 105 °C to deactivate enzymes. Roots and leaf were washed by enough deionized water to remove pollutant attached on surface, oven to consistent weight, grounded to pass a 2-mm steel mesh, and preserved at − 4 °C before chemical analysis. Rice grains were oven to consistent weight, weighed, and then preserved before THg determination.

Soil samples were collected exclusively at the initiation and conclusion of pot experiments, with no intermediate sampling to minimize disturbance to rice root systems. Soils for 16*s rRNA*, *hgcA* and *hgcB* analysis were freeze at − 80 °C. Other soils were air-dried, sieved, and preserved at − 4 °C for chemical determination.

2.3. Chemical analysis of soil properties, Hg and MeHg

Total Hg and methylmercury (MeHg) in soil and rice tissues were analyzed as described in previously published procedures [18]. In brief, about 0.500–1.000 g soil or rice samples were digested in parallel using 5.0 mL of HNO_3 and 1.0 mL of H_2SO_4. THg concentrations were determined by an inductively coupled plasma mass spectrometer (ICP-MS 7500, Thermoelemental, USA). MeHg in soils were extracted by HCl (6 N, soil: HCl solution 1:10 w/v) and then transferred into CH_2Cl_2 phase, separated the organic phase and aerated by pure N_2 at 60℃ water bath to remove CH_2Cl_2, oxidized by adding brominating agent (containing an equal volume of KBr and $KBrO_3$), reduced by $SnCl_2$ (20 % m/v) to transfer Hg^{2+} to elemental Hg, and determined by a CVAFS instrument (Tekran 2600, Canada). THg in soils, rice grains, roots and leaves were labeled as THg, Hgrr, Hgrt, and Hgl, respectively. Analytical accuracy and precision were validated by comparing measured results and certification reference materials, and the recovery of Hg and MeHg varied in the range of 96–103 % and 87–94 %, respectively.

Soil organic carbon (SOC), total nitrogen (TN), and total sulfur (S) were determined in parallel using an Elemental analyzer (Vario EL III, Germany). The dissolved organic matter in soils (DOM) was determined by a TOC analyzer (Gangdong, Tianjin) after extracting DOM by mixing 1.00 g soil with 10.0 mL deionized water. Total phosphorus (P) was determined by colorimetric analysis after digestion in HNO_3–$HClO_4$ solution.

2.4. 16s rRNA and Hg methylation genes (hgcA and hgcB)

High-quality genomic DNA from 0.25 g soil was extracted by the E.Z.N.A.® Soil DNA Kit (Omega Bio-tek, Norcross, GA, U.S.), according to the manufacturer's protocol. We purified the total DNA using 2 % agarose gel electrophoresis and an Fluorometer Qubit 4.0 (Thermo Fisher Scientific, USA). The purified PCR products were subjected to library preparation using the NEXTFLEX Rapid DNA-Seq Kit through the following procedures: (1) Adapter ligation was performed to integrate sequencing adapters; (2) Magnetic bead-based purification was employed to eliminate self-ligated adapter fragments; (3) PCR amplification was conducted to enrich library templates; (4) The PCR products were subsequently purified via magnetic bead-based recovery to generate the final sequencing library. High-throughput sequencing was executed on the Illumina NextSeq 2000 platform (Shanghai Meiji Biomedical Biotechnology Co., Ltd., China).

For Hg methylation genes analysis, in brief, *hgcA*-F (GGNRTYAAYRTNTGGTGYGC) and *hgcB*-R (CADGCNCCRCAYTC-VATRCA) were used to produce 650-bp PCR products as the target fragment [19]. The amino acid sequences of non-redundant gene sets were aligned against the *KEGG* database using DIAMOND (version 0.8.35; http://www.diamondsearch.org/index.php) with BLASTP parameters configured at an expectation value (e-value) cutoff of 1×10^{-5}. Gene-associated KEGG functional annotations were subsequently extracted, and targeted identification and quantification of *hgcA* and

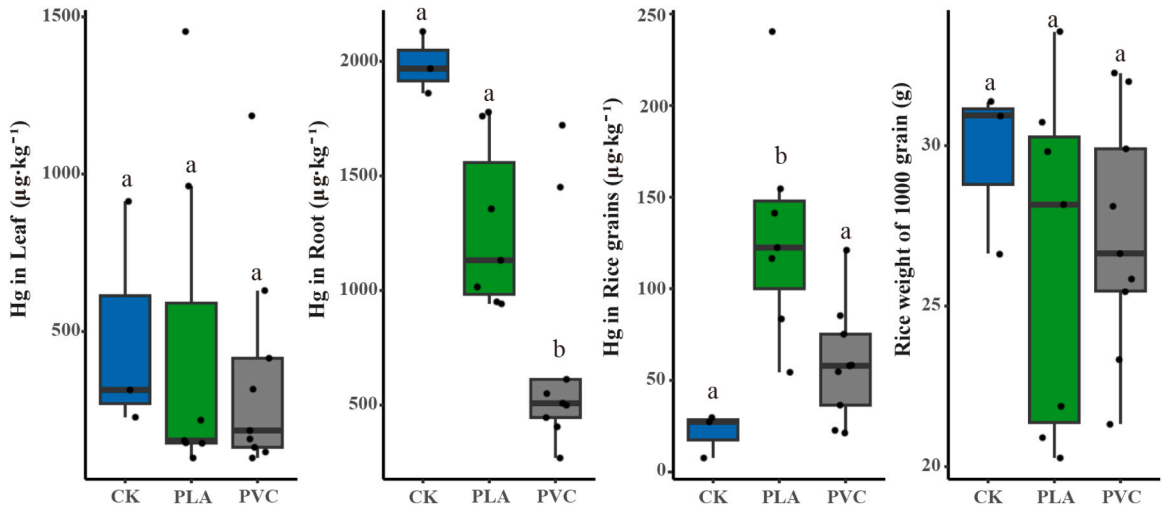


Fig. 2. Comparison of Hg in rice organs and grain weight across CK, PLA and PVC treatments. Multi-comparison results were showed as letters in box plots. The same letters meant no significant difference while the different letters indicated significant difference at $p < 0.05$ level.

hgcB genes were systematically performed.

Raw FASTQ files were de-multiplexed using an in-house perl script, and then quality-filtered by fastp version 0.19.6 and merged by FLASH version 1.2.7 [3] with the following criteria: (i) the reads were truncated at any site receiving an average quality score of < 20 over a 10 bp sliding window, and the truncated reads shorter than 50 bp were discarded, reads containing ambiguous characters were also discarded; (ii) only overlapping sequences longer than 10 bp were assembled according to their overlapped sequence. The maximum mismatch ratio of overlap region is 0.2. Reads that could not be assembled were discarded; (iii) Samples were distinguished according to the barcode and primers, and the sequence direction was adjusted, exact barcode matching, 2 nucleotide mismatch in primer matching. Then the optimized sequences were clustered into operational taxonomic units (OTUs) using UPARSE 11.0.667 with 97 % sequence similarity level. To minimize the effects of sequencing depth on alpha and beta diversity measure, the number of 16S rRNA gene sequences from each sample were rarefied to 20,000, which still yielded an average Good's coverage of 99.09 %, respectively. The metagenomic function was predicted by PICRUSt2 (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States) based on OTU representative sequences. The details for 16*s rRNA* and Hg methylation genes (*hgcA*, *hgcB*) analysis could be found in instruction book of Meiji Bio Company, which can be reached at https://www.majorbio.com/.

2.5. Statistical analysis

All statistical analyses were conducted using R software. One-way analysis of variance (ANOVA) was employed to compare soil physicochemical properties, THg, MeHg, microbial diversity indices, and Hg methylation genes express (*hgcA*, *hgcB*) among CK, PVC, and PLA treatments. Differences in microbial community composition across treatments were assessed using permutational multivariate analysis of variance (PERMANOVA) based on Bray-Curtis dissimilarity [20]. Potential relationships among Hg species (THg, MeHg), soil properties, and microbial parameters were examined using Pearson correlation and linear regression analyses, with significance evaluated at $p < 0.05$. The relative importance of environmental factors influencing THg and MeHg concentrations was determined and ranked using random forest analysis.

Table 1
Ratios of Hg among rice organs across CK, PLA and PVC treatment.

Ratio*	CK	PLA	PVC	*p*
Rice grain/Soil (BAFrice)	0.01	0.10	0.08	0.127
Roots/Soil (BAFroot)	1.59	0.94	0.88	0.204
Leave/Soil (BAFleave)	0.35	0.26	0.43	0.623
Rice grain/Roots (TFrice)	0.24	0.30	0.73	0.361
Leave/Roots (TFl)	0.01	0.11	0.10	0.043
Rice grain/Leave (TFrl)	0.05	0.78	0.26	0.017

* Ratios of THg in rice organs to soil Hg contents were defined as bioaccumulation factors (BAF). Ratios among THg contents in rice organs were defined as transferring factors (TF).

3. Results

3.1. Changes of THg, MeHg and soil properties

MPs significantly altered THg, MeHg, %MeHg, and soil properties. The highest concentrations THg, MeHg, SOC, CN, DOC, and NH_4-N were observed in the PLA treatment, the highest TN, NO_3-N, and pH were found in CK treatments, and the highest %MeHg, sulfur and TP concentrations were found in the PVC treatment. Significant differences were tested for THg between PVC and PLA ($p = 0.006$), for sulfur between PVC and CK ($p = 0.027$), between PVC and PLA ($p = 0.021$) (Fig. 1).

3.2. Accumulation of Hg in rice organs and grains

Mercury accumulation in rice organs exhibited significant variation among the CK, PVC and PLA treatments. The highest THg in rice grains (Hgrr) was observed in the PLA treatment with a mean of 130.46 $\mu g \bullet kg^{-1}$. This value was approximately six-folds of that in CK treatment. Significant differences were detected between PLA and CK ($p = 0.005$) and between PVC and PLA ($p = 0.01$). THg in rice roots (Hgrt) followed the order of CK>PLA>PVC. Significant differences were observed between PVC and CK ($p = 0.001$) and between PVC and PLA ($p = 0.04$). The Hgrt in the CK treatment was approximately 2.76 that of the PVC treatment. THg in leaves (Hgl) showed minimal variation across the treatments. No significant differences were observed in the 1000-grain weight among the treatments. Average weight was 29.64 g (CK), 26.47 g (PLA), and 27.20 g (PVC). These results indicated that MPs addition enhanced Hg accumulation in rice grains, reduced mercury adsorption by roots, and resulted in a decrease in grain yield (Fig. 2).

Comparison of THg ratios between rice organs and soils revealed that MPs addition inhibited Hg adsorption by roots but enhanced accumulation Hg in rice grain. However, the bioaccumulation factors (BAFs) for Hg showed no significant differences across treatments. Notably, the

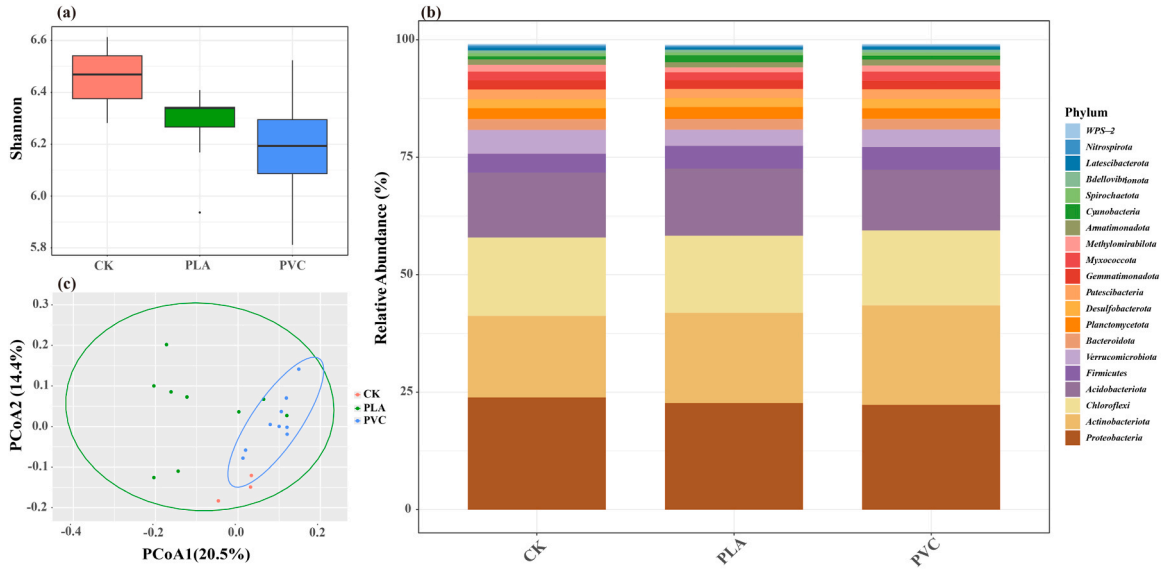


Fig. 3. Microbial abundance and community structure change among CK, PLA and PVC treatment. The Shannon index was used to show alpha diversity change (a), and the same letter meant there was no significant for Shannon diversity among different treatment. Only the most 20 abundant phyla were listed (b). The adjusted principal coordinates analysis based on Bray-Curtis distance (PCoA) method was employed to compare differences in community structure across CK, PLA and PVC treatment (c), and the ellipse overlapped meant MPs did not have significant effects in changing microbial community structure, this was consistent with results of Shannon diversity.

translocation of Hg from roots to grains and from leaves to grains was 0.78 in the PLA, compared to 0.26 in PLA. These findings indicate that the elevated grain Hg concentration observed in the PLA treatment was primarily attributed to the efficient translocation of Hg from leaves to grains (Table 1).

3.3. Microbial communities

MPs addition reduced bacterial alpha diversity. The average Shannon index was 6.454, 6.280, and 6.19 for the CK, PVC and PLA treatments, respectively. However, no significant differences were detected across these treatments ($F = 2.032$, $p = 0.16$) (Fig. 3a). Furthermore, MPs addition had little impact on the bacterial community structure at the phylum level (Fig. 3c).

Bacterial communities exhibited similar compositional profiles across all treatments. *Proteobacteria*, *Actinobacteriota*, *Chloroflexi*, and *Acidobacteriota* were the dominant phyla, collectively accounting for 71.74 %, 72.61, and 72.37 % of the total bacterial abundance in the CK, PLA and PVC treatments. However, bacterial responses to MPs addition were taxon-specific, with the abundance of most phyla decreasing after MPs addition. Notably, the abundances of *Actinobacteriota*, *Firmicutes*, *Desulfobacterota*, *Cyanobacteria*, and *Bdellovibrionota* increased following MPs addition, of which *Actinobacteriota* was the most sensitive, increasing by 1.82 % in the PLA treatment and by 3.82 % in the PVC treatment relative to treatment. Conversely, *Proteobacteria* abundance decreased by 1.18 % (PLA) and by 1.57 % (PVC) compared to CK

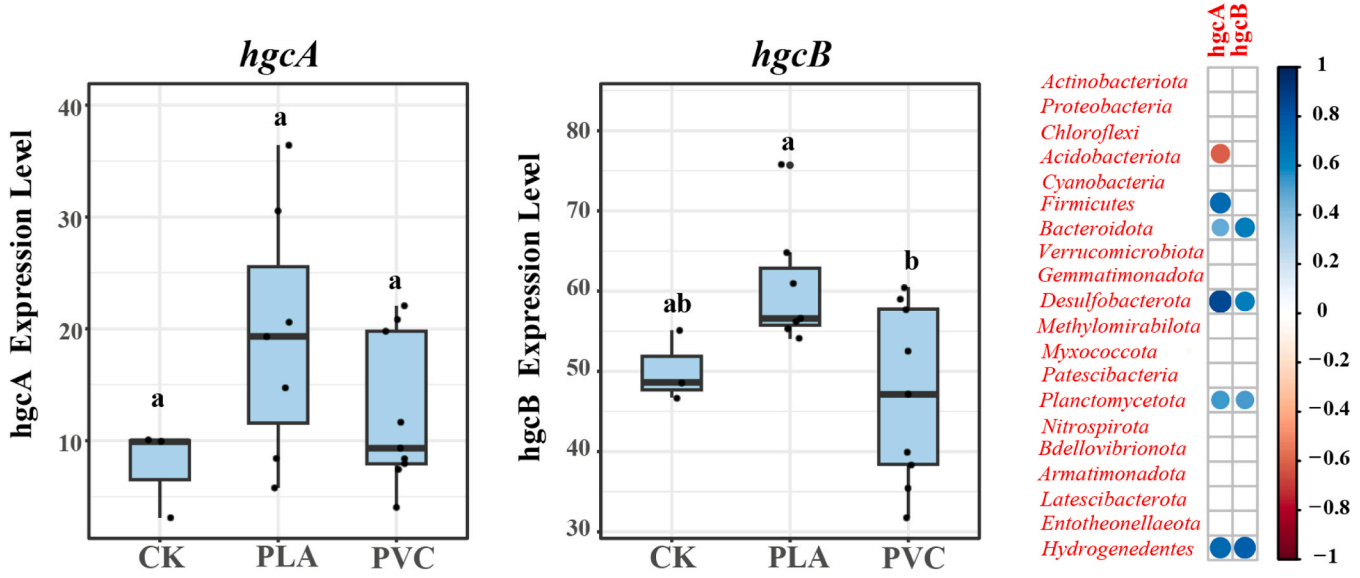


Fig. 4. Expression of Hg methylation genes (*hgcA* and *hgcB*) under CK, PLA and PVC treatments, and correlations between Hg methylation genes and microbial abundance. The same letters meant *hgcA* expression did not differ significantly across CK, PLA and PVC treatment, while *hgcB* expression differed significantly between PLA and PVC treatment. Only correlations at $p < 0.05$ level was presented, and the blanks meant correlations at $p > 0.05$ level.

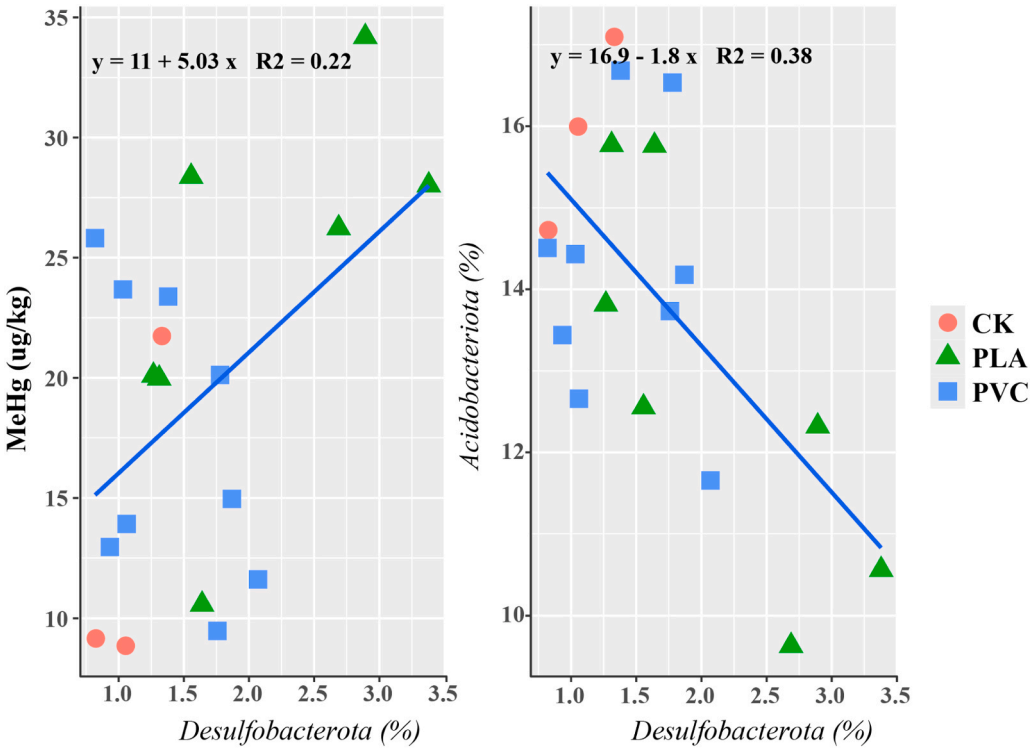


Fig. 5. Relations between *Desulfobacterota* abundance and MeHg, *Acidobacteriota* abundance. Different treatments were labeled by color and shapes.

treatment. *Acidobacteriota* and *Planctomycetota* abundances increased by 0.48 % and 0.22 % in PLA, respectively, but decreased by 0.88 % and 0.11 % in PVC treatments relative to CK treatments. PLA addition reduced *Myxococcota* and *Armatimonadota* abundances, whereas PVC addition promoted them (Fig. 3b).

3.4. Hg methylation genes (hgcA and hgcB) variation under MPs addition

MPs addition significantly promoted Hg methylation genes expression levels. Specifically, *hgcA* expression increased by 152 % under PLA addition and by 61 % under PVC addition, compared to the CK treatment. *HgcB* expression increased by 20.62 % in the PLA treatment but decreased by 6.42 % in the PVC treatment, relative to CK treatment. Significant differences in hgcB expression levels were observed between the PLA and PVC treatments (Fig. 4).

Correlation analysis indicated that the abundances of *Bacteroidota*, *Desulfobacterota*, *Planctomycetota*, and *Hydrogenedentes* were significantly and positively correlated with *hgcA* and *hgcB*. In contrast, *Acidobacteriota* was negatively correlated with *hgcA*, whereas *Firmicutes* was positively correlated with *hgcA*. The strongest positive correlations were observed between *Desulfobacterota* abundance and *hgcA* ($r = 0.851$, $p < 0.001$), and between *Hydrogenedentes* and *hgcB* ($r = 0.768$, $p = 0.001$) (Fig. 4).

4. Discussion

The disintegration of PLA is considered environmentally friendly with complete mineralization potentially, as it does not create persistent MPs in environment. Consequently, PLA is recognized as a viable large-scale substitute for PVC [21]. However, in this study, we observed for the first time that higher MeHg and Hg contents in soils, as well as increased Hg in rice grains under PLA-MPs treatment compared to PVC addition. This implies that PLA remarkably promoted MeHg formation, reduced Hg loss, and thereby prolonged the retention time of Hg in paddy soils. Compared to PVC, PLA addition significantly promoted soil organic matter, the ratios of SOC to nitrogen (CN), DOM, NH_4-N, while mitigating the loss of NO_3-N and TN, and accelerating P loss. In contrast, PVC addition remarkably increased sulfate contents in soils. Notably, PLA addition significantly reduced soil pH. These findings were consistent with previous work [22]. MPs addition changed relative abundance of dominant bacterial taxa, which could affect MeHg formation. Soil pH and *Acidobacteriota* were key factors controlling MeHg formation in the PLA treatment, while *Acidobacteriota*, *Cyanobacteria*, and alpha diversity were the main drivers of MeHg in PVC treatments (Fig. S2). Intriguingly, no significant correlations were observed between *Desulfobacterota* and soil pH. Soil sulfate bacteria prefer an environment with pH of 7.5–8.0 [23]. Although soil pH decreased significantly in the PLA treatment, potentially stimulating *Desulfobacterota* growth, this stimulatory effect may not have reached statistical significance because the mean soil pH remained above 8 in PLA-amended soils. Meanwhile, *hgcA* expression and soil pH were negatively correlated ($r = -0.479$, $p = 0.028$), indicating the lower pH fueled MeHg production by facilitating *hgcA* expression.

Complete degradation of PLA is partly abiotically driven [24], typically requiring several months. As ester bonds are broken down, lactic acid is released and accumulates, causing soil acidification [25]. This acidification could effectively increase MeHg production in freshwater ecosystems [26]. Furthermore, PLA-amended soils yield more DOM than PVC-amended soils. High DOM content in soil plays a critical regulatory role in Hg methylation in paddy soils through two primary mechanisms. First, DOM in PLA-amended soils, likely dominated by low-molecular-weight organic acids, can enhance microbial activity and promote Hg methylation [26]. Second, elevated DOM concentrations strengthen the complexation of Hg^{2+} with DOM [27], limiting its reduction to volatile Hg^0 and thereby reducing Hg loss from soils [28]. This process retains more Hg^{2+} as substrates for methylation. Consistent with this proposed mechanism, PLA-amended soils exhibited higher MeHg concentrations, lower %MeHg, and elevated THg levels compared to the PVC treatment.

Acidobacteriota is a phylum of Gram-negative bacteria and is found in a variety of environments [29]. RF analysis indicated that *Acidobacteriota* plays an important role in controlling Hg methylation in both

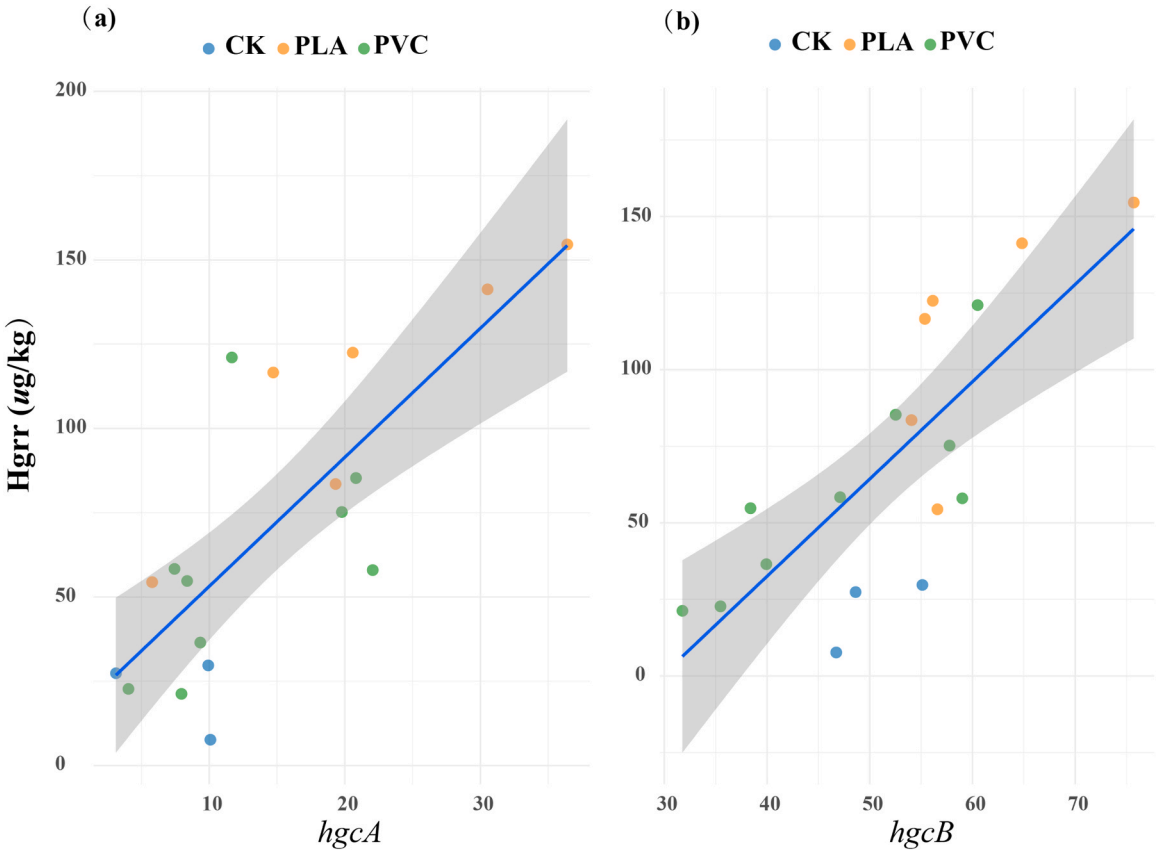


Fig. 6. Correlations between THg in rice grain (Hgrr) and *hgcA*, *hgcB*. Samples under different treatment were labeled by different color.

PLA and PVC treatments (Fig. S2). However, much remains unknown about the role of *Acidobacteriota* in Hg methylation. *Acidobacteriota* might directly catalyze the methylation process as it represented the second most abundant phylum harboring dissimilatory sulfite reductase (*dsrB*) genes (averaging 13 %) [30]. This assumption was further supported by a significant negative correlation, indicating a competitive relationship, between *Acidobacteriota* and *Desulfobacterota* (Fig. 5).

Cyanobacteria were also the main controlling factor for MeHg in the PVC treatment. As *Cyanobacteria* possess neither *hgcA* nor *hgcB*, it was hypothesized that they affect MeHg indirectly, either by creating an anaerobic environment [31], or by promoting sulfate-reducing bacteria [32]. *Desulfobacterota,* which usually serve as the primary methylators [33], effectively promoted MeHg production (Fig. 5). PLA addition increased *Desulfobacterota* abundance to 2.10 % from 1.07 % in the CK treatment, while this value was 1.24 % in the PVC treatment. However, contributions of *Desulfobacterota* on MeHg production might be overlapped and controlled by *Acidobacteriota*, which accounted for 12.91 % and 14.20 % of the total bacterial community in PLA and PVC, respectively (Fig. 3). The lower *Acidobacteriota* abundance in PLA corresponded to more effective MeHg formation by *Desulfobacterota*, a finding supported by the highest correlation coefficients ($r = 0.851$, $p < 0.001$) between *Desulfobacterota* and *hgcA* (Fig. 4).

The *Hydrogenedentes* abundance also showed a positively correlation with MeHg, indicating a promotion effect on MeHg production. *Hydrogenedentes* were putative organic carbon degraders, effective in decomposing lipids and glycerols [34], and their abundance also correlated with *hgcA* ($r = 0.717$, $p = 0.001$) and *hgcB* ($r = 0.768$, $p < 0.001$) expression (Fig. 4). However, neither *hgcA* nor *hgcB* showed significant correlations with soil MeHg or %MeHg.

MPs act as carriers, exhibiting divers interactive effects to metal bioavailability and mobility, and thereby, affecting plant uptake [35, 36]. This study provided the first report demonstrating the differences between PLA and PVC on Hg regarding Hg uptake and bioaccumulation in rice organs. We found that PLA significantly enhanced Hg levels in rice grain while reducing Hg accumulation in roots and leaves. PVC was more effective than PLA in reducing Hg uptake by rice roots, potentially attributed to differences of adsorption and desorption between the two plastics. Although PLA adsorbed Hg^{2+} more quickly and had a higher adsorption capacity (Q_{max} 11.4 $\mu g \bullet g^{-1}$ for Hg^{2+}) than PVC (Q_{max}, 8.7 $\mu g \bullet g^{-1}$ for Hg^{2+}), Hg desorption from PLA occurred more rapidly than from PVC [37]. Though MPs contents used for pot experiments were much higher the real-world agricultural soil, considering the widespread use of PLA plastic products, the risk of Hg accumulation in rice grains due to MPs in paddy fields cannot be overlooked in the future.

MPs could strongly form the plastisphere that comprise the microbial community on plastic debris [38], which acts as the hotpots for elemental transport and microbial habitats. PLA is biodegradable, and consequently, more Hg is released during its mineralization, whereas PVC is not easily degraded. Considering that Hg uptake process by rice roots competes with Hg adsorption by MPs, the mineralization poses a more important constraint on decomposition. Furthermore, Hg accumulation in rice root was greatly hindered by higher sulfur contents, with a significantly negative correlation observed between these parameters (r = -0.584, $p = 0.009$, Fig. S3). Under flooded anaerobic conditions, sulfate is reduced and reacts with Hg^{2+} to form HgS, thereby inhibiting Hg uptake by rice root.

Notably, Hgrr was the highest while Hgrl was the lowest in PLA treatment. The transfer index of Hg from leave to rice grain (TFrl) was also highest in PLA treatment, 0.78, which was three times higher than in the PVC treatment, and 15.6 time higher than in the CK treatment. This indicated that MPs promoted the transfer efficiency of Hg to rice grain. This phenomenon may be ascribed to the relatively high MeHg contents associated with PLA. Due to its lipid solubility, MeHg can easily cross cell membranes and enter grains when transport photosynthates

such as sucrose [39]. This assumption was partly supported by significant positive correlations between Hgrr and *hgcA* ($r = 0.476, p = 0.039$) and between Hgrr and *hgcB* ($r = 0.693$, $p = 0.01$) (Fig. 6). However, since MeHg levels in rice organs were not determined in the present work, and this hypothesis requires further experimental validation.

5. Conclusions

MPs in agricultural soils are growing pollution concern. Biodegradable plastics, represented by PLA, are considered effective substitutes for reducing MPs pollution. Our data align with the emerging consensus that MPs addition may promote MeHg production in soil and prolongs Hg retention in paddy soils, with PLA proving more effective in fueling MeHg formation than PVC. PLA addition significantly altered soil pH. *Acidobacteriota* emerged as the key phylum controlling MeHg in both PLA and PVC treatments. Soil pH and microbes primarily regulate soil MeHg through indirect effects, particularly by modulating Hg methylation processes via competitive interactions with *Desulfobacterota*.

MPs addition increased Hg accumulation in rice organs. Notably, we provided the first evidence that PLA application significantly increase Hg accumulation in rice grain, while PVC addition reduced Hg uptake by rice roots. Given the lower Hg contents in rice leaves and roots under MPs addition, MPs dramatically enhance Hg transfer efficiency to grains. PLA was the most effective in facilitating Hg translocation from leaves to grains. However, the specific roles of PLA and PVC polymers in this translocation process remain unclear. Future studies employing Hg isotope labeling could elucidate Hg transport dynamics within the soil-plant system and reveal allocation mechanisms of Hg to rice organs under MPs exposure.

PLA plastics are increasingly adopted in agriculture as a biodegradable alternative to PVC, aiming to address the persistent issue of plastic residues in soil. However, we contend that extensive replacement of PVC with PLA in Hg-rich agricultural soils warrants cautious consideration. PLA exhibits higher susceptibility to fragmentation into MPs than PVC. Once introduced into Hg-polluted paddy systems, PLA-derived MPs significantly enhance mercury accumulation in rice grains, thereby amplifying health risks associated with dietary mercury exposure.

Environmental Implications

Microplastics (MPs), as emerging pollutants, have been accumulating in historically Hg-polluted paddy fields. The replacement of PVC with biodegradable PLA has become a crucial strategy for addressing MPs pollution, yet its impacts on mercury methylation in soils, and subsequent accumulation in rice organs remain poorly understood. Our results highlight that PLA application could remarkably promote Hg methylation and exacerbate Hg accumulation in rice grains. Our findings emphasize the necessity for comprehensive risk assessment when implementing biodegradable MPs in Hg-rich agricultural soils, particularly regarding food safety implications through enhanced grain Hg enrichment.

CRediT authorship contribution statement

Wu Haitao: Methodology, Conceptualization. **zhang zhongsheng:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Sang Luan:** Methodology, Investigation. **Wu Haobo:** Visualization. **Zhang Zimo:** Methodology, Investigation. **Zhao Wenwen:** Validation, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The present work is supported by the Strategic Priority Research Program of Chinese Academy of Sciences (Grant no. XDA28020401).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2025.140146.

Data availability

Data will be made available on request.

References

[1] Zhao, H., Yan, H., Zhang, L., Sun, G., Li, P., Feng, X., 2019. Mercury contents in rice and potential health risks across China. Environ Int 126, 406–412. https://doi.org/10.1016/j.envint.2019.02.055.
[2] Li, Z., Wang, T., Yang, X., Wen, X., Chen, W., He, Y., et al., 2023. Microbial community function and methylmercury production in oxygen-limited paddy soil. Ecotoxicol Environ Saf 266, 115585. https://doi.org/10.1016/j.ecoenv.2023.115585.
[3] Amin, S., Khan, S., Sarwar, T., Nawab, J., Khan, M.A., 2021. Mercury methylation and its accumulation in rice and paddy soil in degraded lands: a critical review. Environ Technol Innov 23, 101638. https://doi.org/10.1016/j.eti.2021.101638.
[4] Wu, Q., Hu, H., Meng, B., Wang, B., Poulain, A.J., Zhang, H., et al., 2020. Methanogenesis is an important process in controlling MeHg concentration in rice paddy soils affected by mining activities. Environ Sci Technol 54 (21), 13517–13526. https://doi.org/10.1021/acs.est.0c00268.
[5] Kedzierski, M., Cirederf-Boulant, D., Palazot, M., Yvin, M., Bruzaud, S., 2023. Continents of plastics: an estimate of the stock of microplastics in agricultural soils. Sci Total Environ 880, 163294. https://doi.org/10.1016/j.scitotenv.2023.163294.
[6] Yang, X., Li, Z., Ma, C., Yang, Z., Wei, J., Wang, T., et al., 2022. Microplastics influence on hg methylation in diverse paddy soils. J Hazard Mater 423, 126895. https://doi.org/10.1016/j.jhazmat.2021.126895.
[7] Kajal, S.S., Thakur, 2024. Coexistence of microplastics and heavy metals in soil: occurrence, transport, key interactions and effect on plants. Environ Res 262, 119960. https://doi.org/10.1016/j.envres.2024.119960.
[8] Bai, Z., Zhang, Y., Cheng, L., Zhou, X., Wang, M., 2023. Nanoplastics pose a greater effect than microplastics in enhancing mercury toxicity to marine copepods. Chemosphere 325, 138371. https://doi.org/10.1016/j.chemosphere.2023.138371.
[9] Bai, Z., He, Y., Hu, G., Cheng, L., Wang, M., 2024. Microplastics at an environmentally relevant dose enhance mercury toxicity in a marine copepod under multigenerational exposure: multi-omics perspective. J Hazard Mater 478, 135529. https://doi.org/10.1016/j.jhazmat.2024.135529.
[10] Amaral-Zettler, L.A., Zettler, E.R., Mincer, T.J., 2020. Ecology of the plastisphere. Nat Rev Microbiol 18 (3), 139–151. https://doi.org/10.1038/s41579-019-0308-0.
[11] Hao, Y.-Y., Liu, H.-W., Zhao, J., Feng, J., Hao, X., Huang, Q., et al., 2023. Plastispheres as hotspots of microbially-driven methylmercury production in paddy soils. J Hazard Mater 457, 131699. https://doi.org/10.1016/j.jhazmat.2023.131699.
[12] Xin, F., Xiao, X., Dong, J., Zhang, G., Zhang, Y., Wu, X., et al., 2020. Large increases of paddy rice area, gross primary production, and grain production in northeast China during 2000–2017. Sci Total Environ 711, 135183. https://doi.org/10.1016/j.scitotenv.2019.135183.
[13] Li, X., Zhang, J., Gong, Y., Yang, S., Ye, M., Yu, X., et al., 2020. Status of mercury accumulation in agricultural soils across China (1976–2016). Ecotoxicol Environ Saf 197, 110564. https://doi.org/10.1016/j.ecoenv.2020.110564.
[14] Zhang, S., Liu, X., Hao, X., Wang, J., Zhang, Y., 2020. Distribution of low-density microplastics in the mollisol farmlands of northeast China. Sci Total Environ 708, 135091. https://doi.org/10.1016/j.scitotenv.2019.135091.
[15] Fang, F., Wang, Q., Li, J., 2004. Urban environmental mercury in changchun, a metropolitan city in northeastern China: source, cycle, and fate. Sci Total Environ 330 (1), 159–170. https://doi.org/10.1016/j.scitotenv.2004.04.006.
[16] Zhang, Y., Wang, P., Wang, L., Sun, G., Zhao, J., Zhang, H., et al., 2015. The influence of facility agriculture production on phthalate esters distribution in black soils of northeast China. Sci Total Environ 506-507, 118–125. https://doi.org/10.1016/j.scitotenv.2014.10.075.
[17] Zhang, Z.S., Wang, Q.C., Zheng, D.M., Zheng, N., Zhang, X.W., Zhang, X.Y., 2008. Mercury biogeochemistry in the soil-plant-insect system in Huludao city. Acta Sci Circumst (10), 2118–2124. https://doi.org/10.13671/j.hjkxxb.2008.10.030.
[18] Zhao, W.W., Xin, Y., Zhang, Z.S., Zhang, X.H., Wu, H.T., 2023. Accelerated Hg loss and increased methylmercury covary with soil organic matter mineralization and dissolved organic matter humification under warming conditions in permafrost marsh, Northeast China. Environ Res 234, 116593. https://doi.org/10.1016/j.envres.2023.116593.
[19] Du, H., Ma, M., Sun, T., Dai, X., Yang, C., Luo, F., et al., 2017. Mercury-methylating genes dsrB and hgcA in soils/sediments of the Three Gorges Reservoir. Environ Sci Pollut R 24 (5), 5001–5011. https://doi.org/10.1007/s11356-016-8213-9.

[20] Shi, Y., Zhang, L., Do, K.-A., Peterson, C.B., Jenq, R.R., 2020. aPCoA: covariate adjusted principal coordinates analysis. Bioinformatics 36 (13), 4099–4101. https://doi.org/10.1093/bioinformatics/btaa276.
[21] Rajeshkumar, G., Arvindh Seshadri, S., Devnani, G.L., Sanjay, M.R., Siengchin, S., Prakash Maran, J., et al., 2021. Environment friendly, renewable and sustainable poly lactic acid (PLA) based natural fiber reinforced composites – a comprehensive review. J Clean Prod 310, 127483. https://doi.org/10.1016/j.jclepro.2021.127483.
[22] Liu, R., Liang, J., Yang, Y., Jiang, H., Tian, X., 2023. Effect of polylactic acid microplastics on soil properties, soil microbials and plant growth. Chemosphere 329, 138504. https://doi.org/10.1016/j.chemosphere.2023.138504.
[23] Hao, O.J., Chen, J.M., Huang, L., Buglass, R.L., 1996. Sulfate-reducing bacteria. Crit Rev Environ Sci Technol 26 (2), 155–187. https://doi.org/10.1080/10643389609388489.
[24] Rüthi, J., Bölsterli, D., Pardi-Comensoli, L., Brunner, I., Frey, B., 2020. The "plastisphere" of biodegradable plastics is characterized by specific microbial taxa of alpine and Arctic soils. Front Environ Sci 8, 562263.
[25] Tokiwa, Y., Calabia, B.P., Ugwu, C.U., Aiba, S., 2009. Biodegradability of plastics. Int J Mol Sci 10 (9), 3722–3742.
[26] Ullrich, S.M., Tanton, T.W., Abdrashitova, S.A., 2001. Mercury in the aquatic environment: a review of factors affecting methylation. Crit Rev Environ Sci Technol 31 (3), 241–293. https://doi.org/10.1080/20016491089226.
[27] Haitzer, M., Aiken, G.R., Ryan, J.N., 2002. Binding of Mercury(II) to dissolved organic matter: the role of the Mercury-to-DOM concentration ratio. Environ Sci Technol 36 (16), 3564–3570. https://doi.org/10.1021/es025699i.
[28] Si, L., Branfireun, B.A., Fierro, J., 2022. Chemical oxidation and reduction pathways of mercury relevant to natural waters: a review. Water 14 (12), 1891.
[29] Huber, K.J., Papendorf, J., Pilke, C., Bunk, B., Spröer, C., Kirstein, S., et al., 2024. Edaphobacter paludis sp. Nov., a new acidophilic representative of the Acidobacteriota isolated from fen soils. Int J Syst Evolut Microbiol 74 (8). https://doi.org/10.1099/ijsem.0.006500.
[30] Flieder, M., Buongiorno, J., Herbold, C.W., Hausmann, B., Rattei, T., Lloyd, K.G., et al., 2021. Novel taxa of acidobacteriota implicated in seafloor sulfur cycling. ISME J 15 (11), 3159–3180. https://doi.org/10.1038/s41396-021-00992-0.
[31] Monperrus, M., Tessier, E., Amouroux, D., Leynaert, A., Huonnic, P., Donard, O.F. X., 2007. Mercury methylation, demethylation and reduction rates in coastal and marine surface waters of the Mediterranean Sea. Mar Chem 107 (1), 49–63. https://doi.org/10.1016/j.marchem.2007.01.018.
[32] Lehnherr, I., St. Louis, V.L., Hintelmann, H., Kirk, J.L., 2011. Methylation of inorganic mercury in polar marine waters. Nat Geosci 4 (5), 298–302. https://doi.org/10.1038/ngeo1134.
[33] Rincón-Tomás, B., Lanzén, A., Sánchez, P., Estupiñán, M., Sanz-Sáez, I., Bilbao, M. E., et al., 2024. Revisiting the mercury cycle in marine sediments: a potential multifaceted role for Desulfobacterota. J Hazard Mater 465, 133120. https://doi.org/10.1016/j.jhazmat.2023.133120.
[34] Momper, L., Aronson, H.S., Amend, J.P., 2018. Genomic description of 'Candidatus Abyssubacteria,'a novel subsurface lineage within the candidate phylum hydrogenedentes. Front Microbiol 9, 1993.
[35] Naqash, N., Prakash, S., Kapoor, D., Singh, R., 2020. Interaction of freshwater microplastics with biota and heavy metals: a review. Environ Chem Lett 18 (6), 1813–1824. https://doi.org/10.1007/s10311-020-01044-3.
[36] Cao, Y., Zhao, M., Ma, X., Song, Y., Zuo, S., Li, H., et al., 2021. A critical review on the interactions of microplastics with heavy metals: mechanism and their combined effect on organisms and humans. Sci Total Environ 788, 147620. https://doi.org/10.1016/j.scitotenv.2021.147620.
[37] Gao, Z., Cizdziel, J.V., Wontor, K., Olubusoye, B.S., 2023. Adsorption/desorption of mercury (II) by artificially weathered microplastics: kinetics, isotherms, and influencing factors. Environ Pollut 337, 122621. https://doi.org/10.1016/j.envpol.2023.122621.
[38] Rillig, M.C., Kim, S.W., Zhu, Y.-G., 2024. The soil plastisphere. Nat Rev Microbiol 22 (2), 64–74. https://doi.org/10.1038/s41579-023-00967-2.
[39] Meng, B., Feng, X., Qiu, G., Liang, P., Li, P., Chen, C., et al., 2011. The process of methylmercury accumulation in rice (Oryza sativa L.). Environ Sci Technol 45 (7), 2711–2717. https://doi.org/10.1021/es103384v.